TASK SPECIFICATION: P5-05 PHASE P5 — SMART PARKING & ALPR COMPUTER VISION GATE

ALPR Gate Access Decision Engine (Hardware Adapter Port)

Git Branch: `feature/P5-05-gate-api` · Target Pull Request: feature/P5-05-gate-api

Core Objective & Technical Scope:
Low-latency webhook API processing vehicle license plates, verifying active permits, and actuating gate barriers via a pluggable adapter.

Priority Level	Medium	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P5-05-gate-api`
Upstream Dependencies	P5-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Define GateHardwarePort interface and implement SimulatedAlprAdapter.
2. Implement GateDecisionService resolving license plates against active bookings.
3. Log 100% of gate entry/exit events in gate_event audit table.
4. Expose /api/v1/gate/scan webhook endpoint with sub-200ms target latency.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `parking/application/Gate*`
- `parking/infra/SimulatedAlprAdapter.java`
- `parking/api/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Unambiguous access decision (ALLOW/DENY) with explicit reason code for every scan.
- Audit log tracks timestamp, plate, confidence score, and decision.

VERIFICATION & TEST COVERAGE STRATEGY

- GateDecisionServiceTest
- GateControllerTest
- PerformanceLatencyTest